

Geology at ICL

Overview

DEGREE MSci / BSc	DURATION 3-4 years	DEPARTMENT Earth Science & Engineering
TYPICAL OFFER AAA (IB 38)	ADMISSIONS TEST None required	INTERVIEW Yes (offline/online)

Imperial's Geology programme is a fieldwork-intensive degree. From Year 1 you're outdoors, mapping real terrain in Dorset and Cornwall. Scotland, Pyrenees, Sardinia, and the Apennines in later years. The department sits within the Royal School of Mines (RSM), which has been formally affiliated with the Geological Society of London since the 19th century and remains one of the most recognised earth science institutions in the world

What makes it different?

Unlike most geology programmes, Imperial emphasises the quantitative side of earth science alongside the observational. You'll do real programming, work with geophysical datasets, and take modules that sit comfortably at the boundary of earth science, physics, and data science.

Streams: Geology vs Earth & Planetary Science (EPS) vs Geophysics

All three streams (Geology, Geophysics, and EPS) share the same modules in Year 1 Term 1. In Term 2, streams begin to diverge slightly: EPS students choose between the Geology or Geophysics track at that point, while Geology and Geophysics each take one different module. The Year 1 field trip is shared by everyone.

Year 2 sees more differences, with distinct module sets and separate fieldtrips. Geology students do a mapping project at the start of summer, while Geophysics students head to Cyprus. The overlap in foundational content stays high through Year 2, but by then you'll know which direction you're heading.

You can still switch streams up to roughly the end of Term 2 in Year 1, as long as you meet the entry requirements for the course you want to move to. Transfers between BSc and MSci are also possible in the early stages. If you're unsure, speak to your personal tutor or senior tutor, these conversations happen all the time and it is more straightforward than it sounds.

Accreditation & careers

The MSci is professionally accredited by the Geological Society of London, and you'll graduate with the Associateship of the Royal School of Mines (ARSM). Graduates go into environmental consultancy, mining, oil & gas, geotechnical engineering, academia, and increasingly data-heavy roles. About 91% of Earth Science & Engineering graduates are in work or further study within 15 months. They have career days and weekly seminars that can help us with this.

A-Levels & Entry Requirements

Typical offer

AAA, this is the minimum entry standard for both the BSc and the MSci.

Required subjects

Imperial requires at least 2 A-levels from the following list:

- Biology
- Chemistry
- Geography
- Geology
- Mathematics
- Physics

Most successful applicants bring Maths + 1 lab science (Chemistry or Physics).

Recommended subjects

While not individually required, **Mathematics** is strongly favoured. **Physics and Chemistry** open up the more quantitative Year 1 modules (like physical process & geochemistry) from day one.

How to Apply

Personal Statement

The personal statement is the most important part of your application before the interview. There is no admissions test for Geology, so your PS and your grades are what get you shortlisted.

Lead with curiosity, not generic passion

Avoid opening with 'I've always been fascinated by rocks' to be honest, very few people are genuinely in love with rocks from birth, and admissions tutors can tell. Instead, open with something specific: a phenomenon you looked into, a moment in a competition or field trip that reframed how you see the earth, a paper you stumbled on. Be specific.

Connect your experiences explicitly to geology

Every experience you mention, a mini chemistry research IA, a summer programme, an extracurricular, needs to be tied back to a geological concept or process. Don't just list what you did, say what it taught you about the earth. For example, my IA was on dissolved oxygen and salinity, connect that to sedimentation and mineral formation. Admissions tutors are reading for scientific thinking, not a CV.

Mention specific geological tools, landscapes, and rock names

This is a small thing that carries real weight: naming actual geology (theodolite, compass, karst terrain, specific rock types) shows you know the discipline, not just the general idea of it. If you've used geological instruments or visited a geological site, name it and say what you learned.

Add a concrete vision, not just interest

Your closing section should do more than say you love geology. It should say why this degree, at this university, connects to something you actually want to do. Name specific modules you plan to take and explain why they matter to your goals. This is what makes the personal statement feel written for Imperial, not a generic geology application.

Admissions Test - None Required

Geology does not use the ESAT (Engineering and Sciences Admissions Test). Unlike Design Engineering or Computer Science, your grades, personal statement, and interview do all the heavy lifting.

The Interview

Format

One-on-one with a member of academic staff, can be offline/online. It's conversational rather than adversarial, they want to know if you can think scientifically, engage with ideas, and handle the quantitative demands of the course.

What they ask

Expect questions that branch off your personal statement, which is why you need to know every academic thing you wrote inside out. They may ask you to explain a concept from your A-levels, think through a geological scenario from first principles, or discuss something you claimed as an interest. Don't memorise answers, follow the logic and show your thinking out loud.

The Course, Module Breakdown (based on 25-26)

The Geology degree has a shared core for all students in Years 1 and 2, then branches into elective-heavy Years 3 and 4. Fieldwork is woven throughout: Dorset + Cornwall (Year 1), Pyrenees + Scotland (Year 2), Sardinia (Year 3), Apennines (Year 4).

Year 1, Foundations (shared core)

Core	Dynamic Earth and Planets
Core	Stratigraphy & Geomaterials
Core	Deforming the Earth
Core	Physical and Surface Processes
Core	Volcanism and Internal Processes
Core	Programming for Geoscientists
Fieldwork	Geology in the Field (Dorset + Cornwall)
Core (Geology stream)	Life over Deep Time
Choice	Chemistry for Geoscientists OR Low Temperature Geochemistry (depends on A-level)
Choice	Mathematics for Geoscientists OR Maths Methods 1 (depends on A-level)

Year 2, Core Science

Core	High Temperature Geochemistry
Core	Pure and Applied Geophysics
Core	Solar System Geoscience

Core	Maps and Structures
Core	Remote Sensing Earth & Planets
Core (Geology)	Igneous & Metamorphic Petrology
Core (Geology)	Sediments and Stratigraphy
Core (Geology)	Palaeontology and Optical Mineralogy
Fieldwork	Rocks & Structures in the Field (Pyrenees + Scotland), 10 ECTS

Year 3, Specialisation

Core	Independent Project, Mapping (15 ECTS)
Core	I-Explore module, cross-faculty elective (5 ECTS)
Choice (pick 3)	Term 1 electives: Continental Tectonics / Climate / Mining Geology & Engineering / Near Surface Seismic Imaging / Advanced Remote Sensing
Choice (pick 4)	Term 2 electives: Palaeobiology · Sea Levels · Tectonics of the Oceans · Arc Magmatic Processes · Geodynamics · Data Science & ML · Contaminant Transport · Mining Environmental Management · and more
Fieldwork	Integrated Advanced Field Geology, Sardinia (compulsory for Geology students)

Year 4, Masters (MSci only)

Core	MSci Research Project (30 ECTS)
Fieldwork	Field Geology of an Active Mountain Belt, Apennines (compulsory for Geology students)
Choice (pick 6)	Electives from pool: Palaeobiology · Geodynamics · Geohazards · Planetary Physics · Magmatic Processes · Geological Reactive Transport · Applied Geomorphology · Meteorites · Minerals Processing · and more

Degree weightings

- MSci: Year 1 (7.5%) / Year 2 (20%) / Year 3 (36.25%) / Year 4 (36.25%)
- BSc: Year 1 (7.5%) / Year 2 (35%) / Year 3 (57.5%)

Student Life

The Royal School of Mines

You'll be based in the RSM Building on the South Kensington campus. The RSM has its own students' union (RSMU), sports clubs, traditions, events (fresher dinners, end of term dinners, pub crawls, socials, wellbeing events, etc)

Fieldwork culture

The fieldwork is the centrepiece of the degree. Year 1 takes you to Dorset + Cornwall; Year 2 to the Pyrenees and Scotland again; Year 3 to Sardinia; Year 4 to the Apennines. These aren't short day trips, you're mapping unfamiliar terrain, working in small teams, and building the kind of observational skill that defines what a geologist actually is. Most students describe the field trips as the highlight of their degree.

Workload & assessment

The degree is exam-heavy in Years 1 and 2 (exams account for roughly 65% and 75% of assessment respectively), shifting toward coursework and project work in Years 3 and 4. Field reports carry real weight. Time management around field seasons and revision is something to develop early.

Accreditation & careers

The MSci is accredited by the Geological Society of London, and completing it reduces the post-graduation experience needed to become a Chartered Geologist (CGeol). Graduates go into environmental consulting, mining, oil & gas, geotechnical firms, academic research, and data-heavy roles. The ARSM associateship is a respected credential in its own right.